AI-Driven Analytics for Energy-Efficient and Cost-Aware Data Center Planning

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I. Background

- Data centers use substantial energy for computing and cooling.
- Operators must balance workload demand, electricity cost, and carbon impact.
- This study combines descriptive analytics and AI to support efficient and cost-aware data center planning.

II. Dataset Overview

- The dataset includes hourly workload, electricity price, server parameters, and ADS cooling-system states.
- These data support integrated analysis of:
  - power demand
  - operating cost
  - carbon emissions
  - cooling performance

III. Methodology

We use a three-step framework.

i. Step 1: Operational Analytics

Let
$W_t = W_t^{(int)} + W_t^{(batch)}$
denote hourly total workload.
We analyze:

- workload demand
- electricity price
- server requirement
- IT and cooling power
- total facility power
- hourly electricity cost
- hourly carbon emissions

ii. Step 2: Cooling-System Modeling

The adsorption cooling system (ADS) is modeled using:

- inputs:
  - generator temperature $T^{(gen)}$
  - condenser temperature $T^{(cond)}$
- outputs:
  - coefficient of performance COP
  - cooling capacity $q^{(cap)}$

iii. Step 3: AI Surrogate Model

We train a Random Forest multi-output model:

$$\left(widehat(COP), widehat\left(q^{(cap)}\right)\right) = f\left(T^{(gen)}, T^{(cond)}\right) \tag{1}$$

to approximate the ADS operating map.

IV. Key Equations

$$ElectricityCost = EnergyConsumptiontimesElectricityPrice \tag{2}$$

$$CarbonEmissions = EnergyConsumptiontimesCarbonIntensity \tag{3}$$

$$R^2 = 1 - frac\left(sum_i\left(y_i - hat(y)_i\right)^2, sum_i(y_i - bar(y))^2\right) \tag{4}$$

V. Workload and Electricity Price

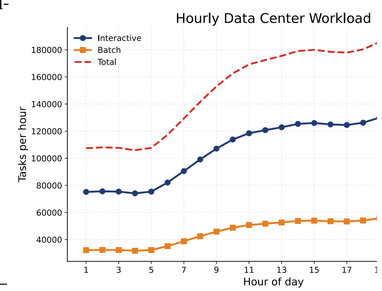

Figure 1. Daily workload pattern.

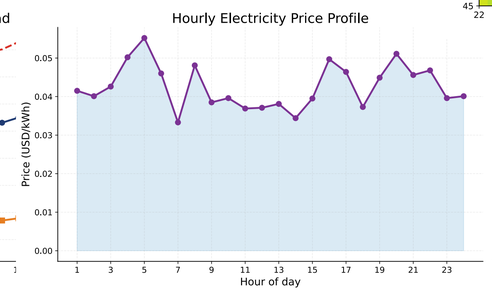

Figure 2. Hourly electricity price.

Key Insight: Workload and price peaks do not fully align, creating opportunities for cost-aware scheduling of flexible demand.

VI. Power, Cost, and Carbon Results

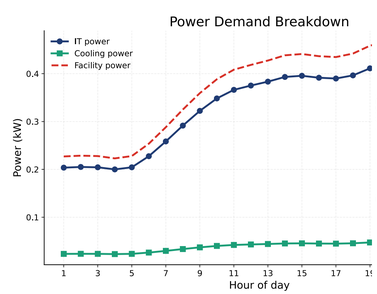

Figure 3. Facility power breakdown.

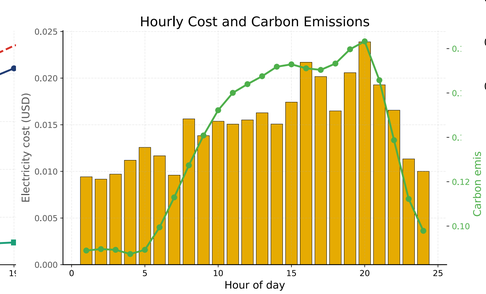

Figure 4. Cost and carbon emissions.

Observation: Higher workload increases both IT and cooling demand, which raises cost and carbon emissions.

VII. Cooling-System Performance Maps

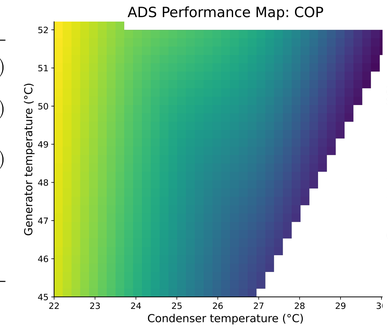

Figure 5. ADS COP map.

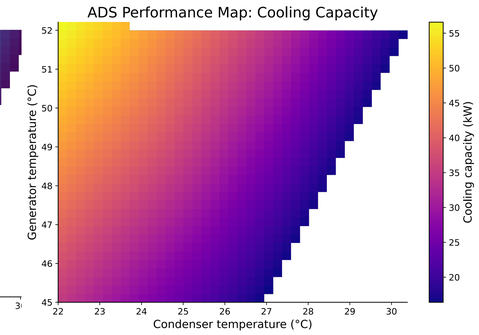

Figure 6. ADS capacity map.

Key Finding: Maximum efficiency and maximum capacity occur in different operating regions.

VIII. AI Model Validation

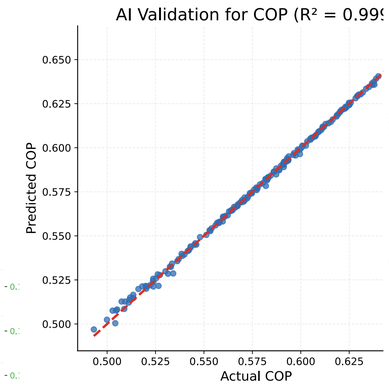

Figure 7. AI validation for COP.

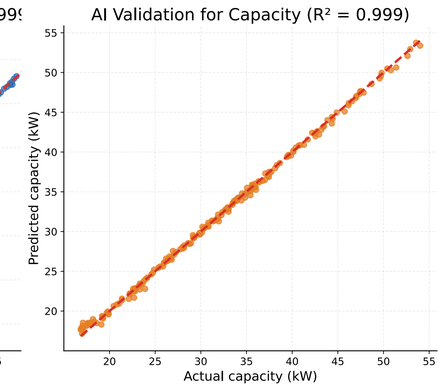

Figure 8. AI validation for capacity.

Model Result: The Random Forest surrogate predicts both COP and cooling capacity accurately and can support optimization and control.

IX. Conclusion

• Workload and price profiles reveal scheduling opportunities • Facility power, cost, and emissions vary substantially over time • ADS performance is strongly nonlinear • A Random Forest surrogate accurately predicts COP and cooling capacity